RAJALAKSHMI ENGINEERING COLLEGE

Akshaya S 251001013-2025-IT

A2

Started on	Tuesday, 25 August 2026, 11:06 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 11:10 AM
Time taken	3 mins 58 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n, c = 0;
    int i, j, k;

    scanf("%d", &n);

    i = n / 2;
    c++;   // outer loop initialization

    while (1)
    {
        c++;   // outer loop condition
        if (i >= n)
            break;

        j = 1;

        while (1)
        {
            c++;   // middle loop condition
            if (j >= n)
                break;

            k = 1;

            while (1)
            {
                c++;   // inner loop condition
                if (k >= n)
                    break;

                c++;   // original c++

                k = k * 2;
            }
```

```
39
40              j = j * 2;
41          }
42
43          i++;
44      }
45
46      printf("%d", c);
47
48      return 0;
49  }
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course